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Statistical modelling of extreme daily rainfall over Aotearoa New Zealand

Suzanne Rosier [a,*], Shalin Shah [b], Greg Bodeker [b], Trevor Carey-Smith [a], David Frame [c,d], Dáithí A. Stone [a]

[a] NIWA, 301 Evans Bay Parade, Wellington, 6021, Aotearoa, New Zealand
[b] Bodeker Scientific, 42 Russell Street, Alexandra, 9320, Aotearoa, New Zealand
[c] Victoria University of Wellington, Kelburn, Wellington, 6012, Aotearoa, New Zealand
[d] University of Canterbury, Private Bag 4800, Christchurch, 8140, Aotearoa, New Zealand

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ABSTRACT

Extreme rainfall in New Zealand, and how best to characterise expected changes in those extremes as the climate warms, is investigated using very large ensembles of regional climate model simulations at five different 'epochs' of climate change (pre-industrial, present-day, and three future states at 1.5 °C, 2.0 °C, and 3.0 °C above pre-industrial). Different constructs of non-stationary Generalised Extreme Value (GEV) models are explored to determine which provides the most accurate estimates of extreme rainfall for the minimum model complexity. The different GEV model constructs vary the number of parameters (location, scale and shape) that are assumed to vary as climate changes, summarised as a linear dependence on Southern Hemisphere mean land surface temperature. Non-stationarity is also explored a different way, with a stationary GEV fitted separately within each of the five 'epochs'. These different models are applied to annual maximum one-day rainfall at eight locations around the country, chosen to be broadly representative of the various rainfall regimes countrywide. In situations with fair but not enormous sample sizes, such as with long historical records, the model in which only the location and scale, but not the shape, parameters vary with warming has the tightest sampling uncertainty without introducing substantial bias. According to this GEV model, 1-in-100-year rainfall increases with warming at all eight locations, ranging from about 5%/°C in most of the country to 8%/°C in the north. The change arises from an increase in the location parameter, with only a proportional increase in the scale parameter, consistent with extreme rainfall increases dictated by anthropogenic increases in specific humidity.

1. Introduction

Due to the steep topography of Aotearoa New Zealand (NZ), landslips, pluvial flooding, and fluvial flooding pose severe hazards for the built and natural environments (Frame et al., 2020; Pastor-Paz et al., 2020). Since these hazards are all driven by extreme rainfall events, regional planning and civil engineering activities require estimates of extreme precipitation to assess current and future risks associated with these hazards. In the presence of anthropogenic climate change, however, empirical statistics from rain gauge observations of past precipitation accumulations cannot be used alone as those observations are limited in sample size and are from a climate era that is unlikely to be representative of the future. Neither though can the other possible data source – the output from climate model simulations – be used alone, not least because current generation models still have large biases in absolute rainfall accumulations (Campbell et al., 2024; Herold et al., 2016; Ackerley et al., 2012). What needs to be done, therefore, is to combine gauge and climate model data in a way that uses the strengths of both: the in situ accuracy of the gauges and the statistical robustness that accrues from the availability of large ensembles of climate model simulations (Bjarke et al., 2024; Tapiador et al., 2017).

One approach is to fit a statistical model to the available data, including global warming covariates in the model's parameters, with some parameters fitted using the climate model data and others using the observed data (Carey-Smith et al., 2018). If the statistical model is of the form $Y \sim P(\alpha_0+\alpha_1 T)$, where T is the covariate (e.g., global mean surface temperature), then the α_0 and α_1 parameters are first estimated through a fit to the output of climate model simulations run through a range of anthropogenic forcings. The α_0 estimate is then discarded, and the statistical model is fit to the observed gauge data, but with α_1 held fixed at the climate-model-based value. Climate models can be run

* Corresponding author.
E-mail addresses: suzanne.rosier@niwa.co.nz (S. Rosier), shalin@bodekerscientific.com (S. Shah), greg@bodekerscientific.com (G. Bodeker), trevor.carey-smith@niwa.co.nz (T. Carey-Smith), david.frame@canterbury.ac.nz (D. Frame), daithi.stone@niwa.co.nz (D.A. Stone).

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many times to generate large samples of the response to anthropogenic forcing, and hence can produce a sufficiently large sample for statistically robust estimation of the covariate parameter (Wehner, 2020). If the climate model data are expressed in normalised form, for instance percent of annual mean precipitation, then we might hope that the estimate of α_1 is decoupled from absolute biases in the climate model precipitation output. On the other hand, gauge data are considered unbiased, and inasmuch as they may in fact contain biases, these are already accounted for in hazard models based on the reference gauges. As such, this method of combining information from the two data sources works to the strengths of both, and in effect bias corrects the climate model data.

For extreme daily precipitation accumulations, we might assume that precipitation events occur frequently throughout the year and that extreme events are produced by a common stochastic process. With these conditions, the distribution of extreme events should conform to the generalised extreme value family of distributions (GEV, Coles, 2001) and the relevant parameters can be estimated through the block maxima approach (e.g. Risser and Wehner, 2017). The GEV distributions depend on three parameters, but if global warming covariates are used to linearly expand each parameter, this doubles the number of parameters to be estimated (Kharin et al., 2007). The accompanying increased uncertainty in the estimate of each parameter leads to the question of whether the global warming covariate is needed for all parameters. Put another way, can we reduce the uncertainty in the statistical description of rainfall extremes without a relative increase in systematic error by removing one, two, or even all of the global warming covariate parameters (Risser and Wehner, 2017)?

There have been many studies addressing this question as they seek to explore the non-stationary nature of rainfall extremes, often motivated by the need to incorporate the effects of climate change into the design of hydrological infrastructure for flood-proofing purposes. Schlef et al. (2023) (their Table 1) present a good overview of various representative studies to date, all of which assume the GEV distribution of extremes and the majority of which choose to fix the shape parameter whilst allowing either the location or scale parameters, or both, to vary. Studies by Jayaweera et al. (2023) and Prosdocimi and Kjeldsen (2021) conclude that optimal results are obtained if both location and scale parameters are allowed to vary together. The shape parameter, which controls the tail of the distribution and is inherently less well sampled (Tyralis et al., 2019), is commonly kept fixed because of a lack of data. Our unusually large datasets prompted us to investigate the full complexity of non-stationarity in all three parameters, and in this our study is similar to that of Brown et al. (2014), who investigated rainfall and temperature extremes using a non-stationary GEV model with all three parameters allowed to vary with global temperature, and who concluded that the shape parameter should remain fixed, mainly because of insufficient data to estimate this parameter adequately.

In this study we have ensured a high degree of statistical robustness by exploiting the very large ensembles of climate model simulations available in the 'weather@home' project for Australia and New Zealand (weather@home/ANZ, Black et al., 2016). These large ensembles are created using computing power donated by thousands of volunteers worldwide. Weather@home/ANZ runs its regional model at ~50 km resolution, thus resolving the main geographic and topological features of NZ. Here, we have exploited this very large number of simulations to:

1. assess the behaviour of different non-stationary GEV models in the NZ setting;
2. examine whether, and under what conditions, any particular GEV model might be optimal insofar as providing the most accurate description of the statistical evolution of extreme precipitation under a changing climate for the minimum model complexity.

The weather@home/ANZ simulations have been run under five different levels of global mean temperature covering the range from pre-industrial to what might plausibly be expected at the end of the 21st century under a business-as-usual greenhouse gas emissions scenario.

2. Methods

2.1. Statistical models

We use the GEV statistical model to describe the distribution of annual maxima of daily precipitation accumulations. The basic GEV model follows:

$$G(z) = \exp\left[-\left\{1+\xi\left(\frac{z-\mu}{\sigma}\right)\right\}^{\frac{1}{\xi}}\right] \tag{1}$$

where μ, σ, and ξ are the location, scale, and shape parameters respectively (Coles, 2001). We set the parameters to have an invariant component and a second component that depends on T'_{SHland}, the annual mean surface temperature anomaly with respect to pre-industrial times (nominally 1765) averaged over Southern Hemisphere land:

$$\mu = \mu_0 + \mu_1 T'_{SHland} \tag{2}$$

$$\sigma = \sigma_0 + \sigma_1 T'_{SHland} \tag{3}$$

$$\xi = \xi_0 + \xi_1 T'_{SHland}. \tag{4}$$

Here, our estimates of T'_{SHland} were taken from the MAGICC (Model for the Assessment of Greenhouse gas Induced Climate Change) model (Meinshausen et al., 2011). Global climate policy is usually measured against global mean temperature, but we use Southern Hemisphere land temperature because it omits the Northern Hemisphere anthropogenic aerosol influence that is largely irrelevant for NZ and because it highlights the polar stratospheric ozone influence that is important for the southern mid latitudes (Banerjee et al., 2020).

While the use of six parameters provides the flexibility to reduce systematic error in the description of extreme rainfall hazard across a range of anthropogenic warming (Kharin et al., 2007), the large number of parameters will induce a large uncertainty in measurement. If we can be confident that one (or more) of the temperature-varying parameters are not needed for a GEV model with sufficiently low systematic error, we could exclude it and thus reduce the uncertainty in the quantification of rainfall hazard (Cooley et al., 2007). To determine which parameters can be excluded, we explore the performance of all possible combinations between just the three invariant parameters ('STAT') and the full six parameters listed in Eqs. (2), (3), and (4). These combinations are listed in Table 1 with one additional model ('EPOCH') in which temperature-invariant μ, σ, and ξ parameters are fitted separately for the five different anthropogenic warming levels and then μ_0 and μ_1 are calculated via a linear fit to those various μ values against T'_{SHland}, and similarly for the scale and shape parameters. There may be physical arguments against the use of some of the temperature-dependent variables; in particular, if increases in extreme precipitation accumulation are expected to scale with moisture while relative humidity remains constant, then both the ξ_1 parameter should be zero and the σ_1 should simply scale in proportion to μ_1 (Wilson and Toumi, 2005). We have included these options though, in case precipitation changes over parts of NZ show more complex behaviour.

2.2. Climate model data

We use daily precipitation output from the weather@home/ANZ (Australia-NZ) project (Black et al., 2016). The HadRM3P regional climate model (Jones et al., 2004) is run at ~50 km resolution over the Australasian CORDEX domain, embedded within the global HadAM3P model (Pope et al., 2000) run at ~150 km resolution. To produce a large sample of daily precipitation time series, the climate model

Table 1
The nine different GEV models examined in this study.

Model name	Description	Parameters
STAT	Stationary GEV	μ_0, σ_0, ξ_0
EPOCH	Separate epochs	μ_0, μ_1, σ_0, σ_1, ξ_0, ξ_1
MU	Expansion in μ only	μ_0, μ_1, σ_0, ξ_0
SIGMA	Expansion in σ only	μ_0, σ_0, σ_1, ξ_0
XI	Expansion in ξ only	μ_0, σ_0, ξ_0, ξ_1
MU-SIGMA	Expansion in μ and σ	μ_0, μ_1, σ_0, σ_1, ξ_0
MU-XI	Expansion in μ and ξ	μ_0, μ_1, σ_0, ξ_0, ξ_1
SIGMA-XI	Expansion in σ and ξ	μ_0, σ_0, σ_1, ξ_0, ξ_1
MU-SIGMA-XI	Expansion in all three parameters	μ_0, μ_1, σ_0, σ_1, ξ_0, ξ_1

combination is run over 12- and 18-month periods using the climate*prediction*.net public distributed computing facility (Allen, 1999), with each simulation starting from a different initial weather state. In this way multi-thousand member ensembles of simulations are created. From both the 12- and 18-month runs, we use data over a 12-month period; this is December to November in the 12-month runs, and the final 12-month period of August to July in the 18-month simulations. The model is run with observed OSTIA (Operational Sea Surface Temperature and Sea Ice Analysis) sea surface temperatures and sea ice concentrations from the 2006 to 2015 period (Donlon et al., 2012).

The weather@home ensembles are run under five different global warming epochs. Each of these differs from the 2006 to 2015 climate by modification of anthropogenic radiative forcings, subtraction/addition of a (spatially and seasonally-varying) sea surface temperature response to those forcings, and a corresponding change in sea ice concentration. While the epochs are defined via the anthropogenic radiative and sea surface forcing, we interpret them via T'_{SHland}, the annual mean Southern Hemisphere mean land temperature anomaly with respect to ca. 1765 (an arbitrary reference date which ensures all T'_{SHland} are positive). The five epochs/experiments are:

1. A "pre-industrial" epoch created by setting anthropogenic radiative forcings to circa year 1850 values and by removing an estimate of the attributable sea surface warming pattern from the observed OSTIA temperatures. In our model, 10 different such SSTs deltas are used based on the attributable warming from various CMIP5 models (Black et al., 2016), with an associated spread of T'_{SHland} varying between 0.06 °C and 0.58 °C.
2. A "present-day" epoch using observed anthropogenic and natural radiative forcings and ocean conditions, with an associated T'_{SHland} varying between 0.75 °C and 1.47 °C.
3. A "1.5 °C" epoch run following the protocols of the HAPPI project (Half a degree Additional warming, Prognosis and Projected Impacts, Mitchell et al., 2017), in which anthropogenic radiative forcings are set to year-2100 values under the RCP2.6 greenhouse gas emissions scenario, and sea surface temperatures are increased according to the warming to the 2090s in a multi-model mean of CMIP5 models driven by the RCP2.6 emissions scenario. Because the land warms more than the ocean, T'_{SHland} is 1.73 °C.
4. A "2.0 °C" epoch run following the protocols of the HAPPI project, in which anthropogenic radiative forcings are set to year-2095 values under a linear combination of the RCP2.6 and RCP4.5 emissions scenarios and the sea surface temperatures are increased according to the warming to the 2090s in a linear interpolation between the multi-model mean of CMIP5 models driven by the RCP2.6 emissions scenario and those models driven by the RCP4.5 emissions scenario. For this epoch, T'_{SHland} is 2.28 °C.
5. A "3.0 °C" epoch run following the expanded protocols of the HAPPI project (Shiogama et al., 2020), in which anthropogenic radiative forcings are set to year-2095 values under a linear combination of RCPs 4.5 and 8.5, and the sea surface temperatures are increased according to the warming to the 2090s in a linear interpolation between the multi-model mean of CMIP5 models driven by the RCP4.5 emissions scenario and those models driven by the RCP8.5 emissions scenario. For this epoch, T'_{SHland} is 3.40 °C.

A summary of the epochs is provided in Table 2. We extract the annual maximum one-day rainfall total (Rx1day) from all of the simulations for the analysis. Calculations are performed separately for random selections of 40, 60, 80, 100, 150, and 200 years of data (and hence maxima), with selection divided evenly across the five epochs (e.g. eight Rx1day values are extracted in each epoch for the generation of a 40-year record for analysis). These 40 to 200 sample sizes were selected to provide an indication of the statistical stability of the results for what might be considered analogous measurement record lengths. In addition, however, for comparison and to maximise the utility of the large datasets, we also include calculations using 2500 years of data. Confidence intervals are estimated using a bootstrap method with resampling 10,000 times. Successful use of the GEV distributional assumption with the large weather@home ensembles has previously been demonstrated by Sippel et al. (2015).

2.3. Geography

The analysis is performed at eight locations around the country, chosen to be broadly representative of the variety of climate regimes found in NZ (Fig. 1). On the North Island, Whangarei (in the Far North) has a subtropical climate, while New Plymouth and Gisborne have mid-latitude climates exposed to westerly and easterly storms respectively (Prince et al., 2021; Reid et al., 2021). The Southern Alps dominate the South Island by forcing uplift of moisture-laden air in the prevailing westerlies (Henderson and Thompson, 1999); Milford Sound, on the West Coast, is extremely wet. Nelson, exposed to the North, receives frequent wet events from atmospheric rivers (Reid et al., 2021), while Christchurch and Dunedin are dry and, like Gisborne, only receive extreme rainfall amounts when flows reverse against the prevailing westerlies (Prince et al., 2021; Stone et al., 2022). Invercargill is exposed to cold mid-latitude westerlies.

2.4. Observational data

We evaluate our model performance by comparison with observations from the New Zealand Virtual Climate Station Network, VCSN (Tait et al., 2006). The VCSN takes observations of daily rainfall from around 600 observing sites nationwide and converts them onto a regular 5 km×5 km grid by use of a thin plate smoothing spline interpolation technique (Hutchinson, 1995). To do this, VCSN is guided by a nationwide climatological (1951–1980) rainfall surface as the basis for the interpolation. Significant fractions of the country are data sparse, in particular high altitude regions of the South Island, which, together with the relatively short length of record, compromises the utility of the climatological rainfall surface to some degree, especially when considering rainfall extremes. Whilst VCSN nevertheless represents a convenient dataset for climate studies, large ensembles of sufficiently high resolution climate model data can constructively augment the information available about New Zealand rainfall extremes.

3. Results

3.1. Climate model evaluation

Quantile–quantile plots comparing the weather@home/ANZ Rx1day precipitation against equivalent values from the NZ VCSN gridded observational product (Virtual Climate Station Network, Tait et al., 2006) are shown in Fig. 1 for each of the eight locations. The plots follow the 1-to-1 diagonal line quite closely except at Milford

Table 2
Summary of the five different climate epochs (experiments) examined in this study.

Epoch name	Anthropogenic radiative forcings	Sea surface temperatures	T'_{SHland}
Pre-industrial	ca. 1850	OSTIA cooled by CMIP5 attributable warming pattern (Black et al., 2016)	0.06–0.58 °C
Present-day	ca. 2010	OSTIA	0.75–1.47 °C
1.5 °C	RCP2.6	OSTIA warmed by HAPPI 1.5 °C pattern (Mitchell et al., 2017)	1.73 °C
2.0 °C	Linear combination of RCPs 2.6 and 4.5	OSTIA warmed by HAPPI 2.0 °C pattern (Mitchell et al., 2017)	2.28 °C
3.0 °C	Linear combination of RCPs 4.5 and 8.5	OSTIA warmed by HAPPI 3.0 °C pattern (Shiogama et al., 2020)	3.40 °C

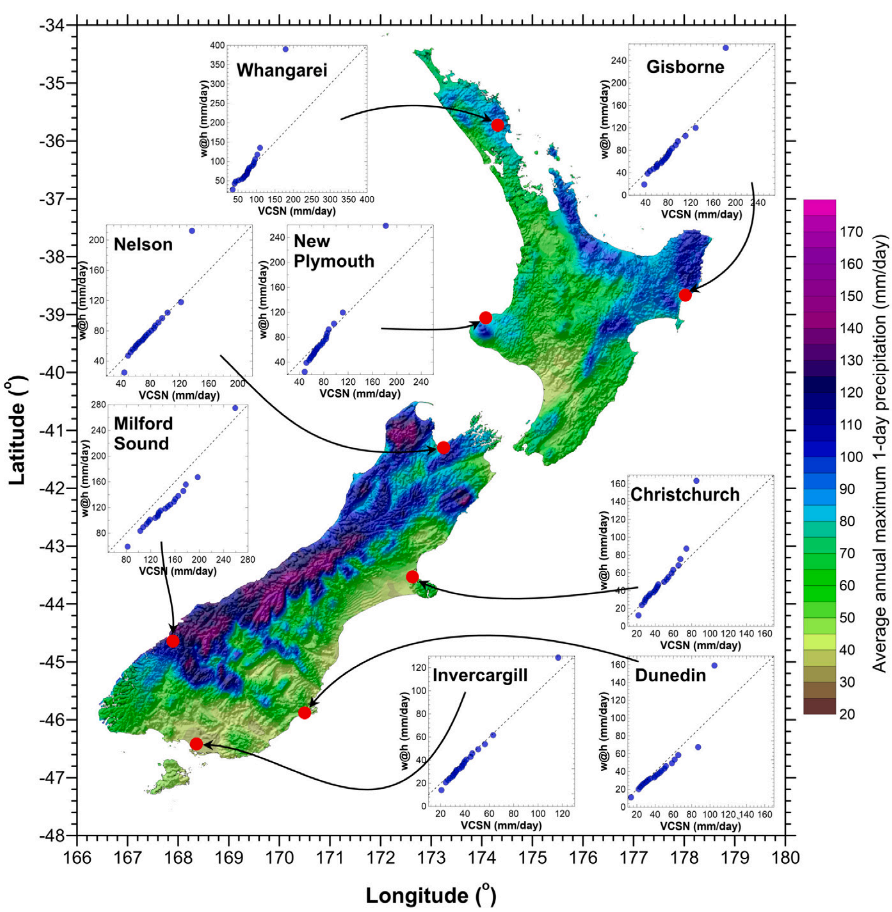

Fig. 1. Map of Aotearoa New Zealand showing the eight representative locations studied in this paper. The background colour map shows a 40-year (1980–2019) climatology of Rx1d values derived from the MSWEP precipitation dataset (Multi-Source Weighted-Ensemble Precipitation (Beck et al., 2017, 2019a,b)), while shading indicates orography. Inset panels show quantile–quantile comparisons, at the eight locations, of weather@home/ANZ Rx1d precipitation against equivalent values from gridded observational data from the NZ Virtual Climate Station Network (VCSN) (Tait et al., 2006). Points are plotted at 21 quantiles (0, 5, 10, …90, 95, 100). Note that in almost all cases the absolute extreme (100th percentile) is substantially larger in the model than observed, as anticipated using such a large model ensemble.

Sound, where weather@home/ANZ does not quite match the exceptionally large observed Rx1day values. The other discrepancy is that in the wettest of the 21 quantiles (0%, 5%, 10% …95%, 100%), the precipitation extremes from weather@home/ANZ exceed those observed, as might be expected given such a large model sample. A two-sample Kolmogorov–Smirnov test (p-level 0.05) revealed that the weather@home/ANZ distribution differs significantly from the VCSN distribution at Milford Sound and New Plymouth, but the comparison passes the similarity test for all stations' fractional anomalies.

3.2. Stability of GEV parameters

Our first task is to determine what the true GEV model is for each location, inasmuch as we can assume that annual maxima in the weather@home/ANZ simulations are behaving like a single stochastic GEV model with linear trends in the parameters. Two of the GEV models are fully comprehensive in allowing for trends in all of their parameters: the EPOCH and MU-SIGMA-XI models.

The estimated values for the fixed terms in the GEV parameters for the EPOCH model are shown in Fig. 2, and for the trend terms in Fig. 3. For this GEV model construct, fixed epoch-dependent terms are estimated separately for each of the five epochs, and the final fixed and trend terms for the GEV parameters are estimated through a linear least squares fit to those epoch-dependent terms as a function of T'_{SHland}. The estimated values of the trend terms are stable as a function of record length (from 40 to 2500 years) for all locations, as are the estimated values of the (fixed) μ_0 and ξ_0 terms. However, the estimates of the fixed term of the scale parameters, σ_0, are lower for short records, rising and gradually converging towards higher values for longer records. Over the 40-year to 2500-year record length range shown in Fig. 2 (and, indeed, over the 40- to 200-year record length range also), the differences amount to about 20 to 30% across the eight locations.

In contrast, the estimated values of all parameters, including of σ_0, are stable across the full range of periods for the MU-SIGMA-XI model (Figs. 4 and 5). In this GEV model construct, the trend terms are incorporated as covariates within the parameters of the GEV model and

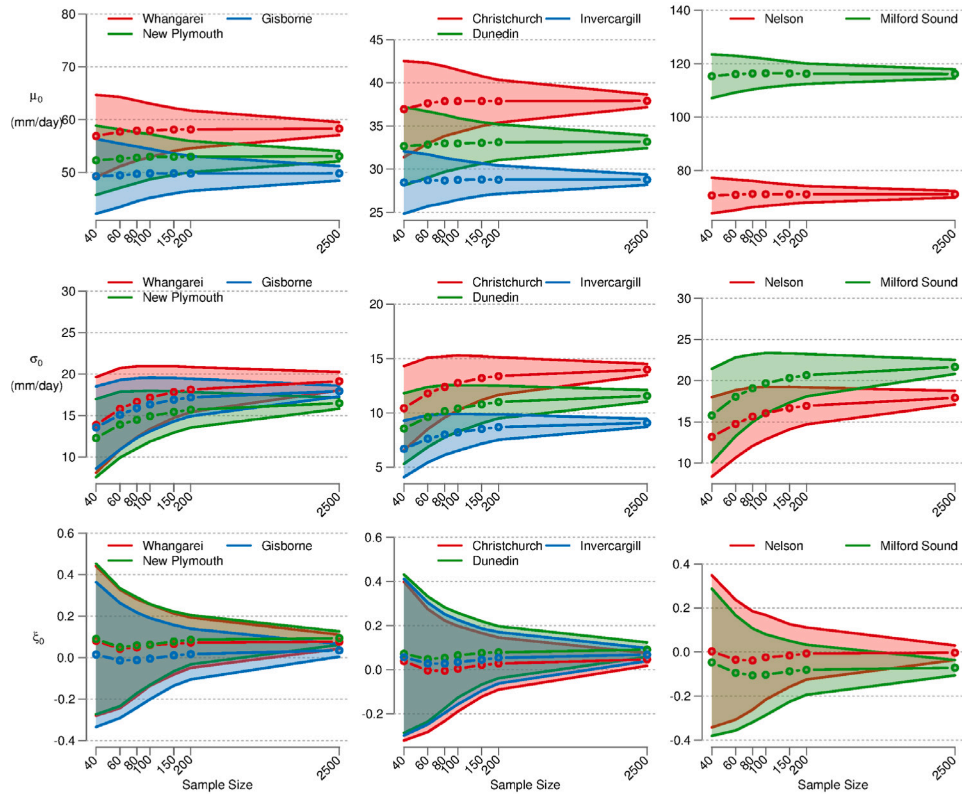

Fig. 2. The three fixed GEV parameters (rows: μ_0, σ_0 and ξ_0) computed for the EPOCH model as a function of sample size (number of years of model data used) and location (colours, and grouped into regions in the columns). The circles indicate the mean estimate and the shading the one sigma standard error.

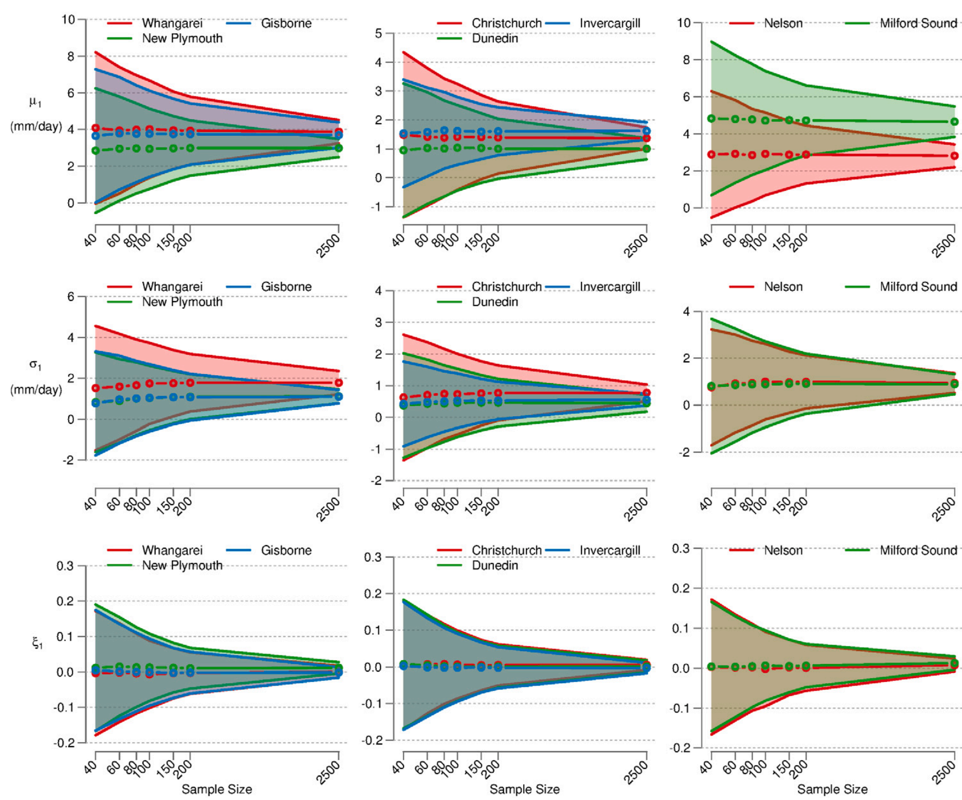

Fig. 3. The three trend GEV parameters (rows: μ_1, σ_1 and ξ_1) computed for the EPOCH model as a function of sample size (number of years of model data used) and location (colours, and grouped into regions in the columns). The circles indicate the mean estimate and the shading one sigma standard error.

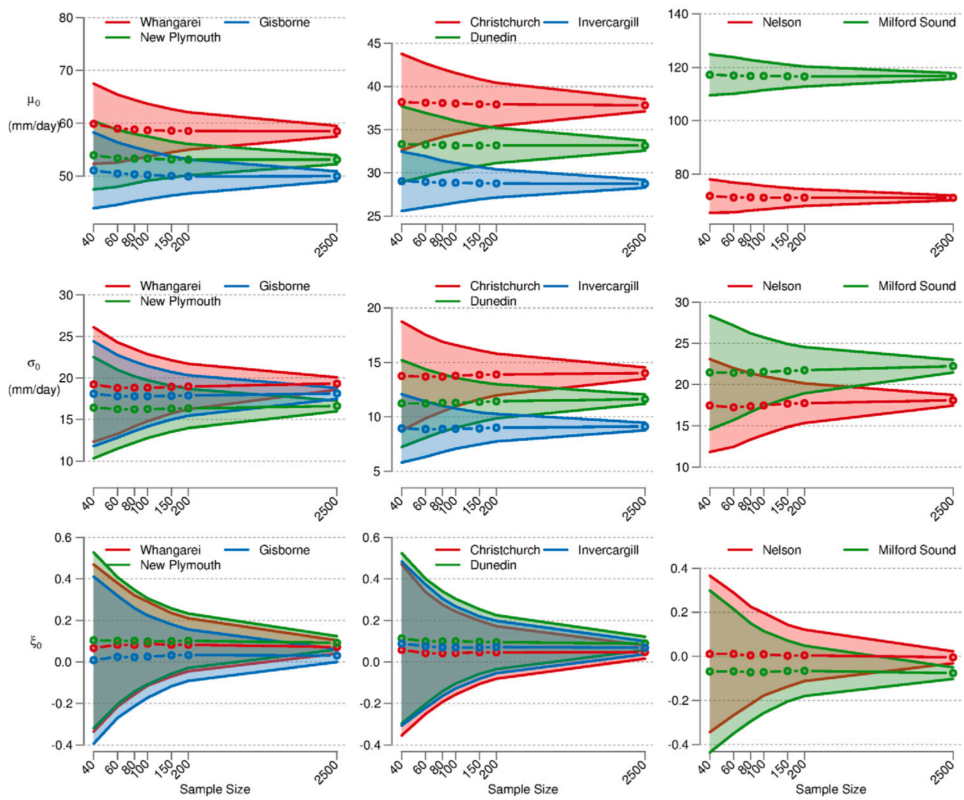

Fig. 4. The three fixed GEV parameters (rows: μ_0, σ_0 and ξ_0) computed for the MU-SIGMA-XI model as a function of sample size (number of years of model data used) and location (colours, and grouped into regions in the columns). The circles indicate the mean estimate and the shading the one sigma standard error.

fit process. The estimated values of the μ_0, ξ_0, μ_1, σ_1, and ξ_1 for 2500-year record lengths are essentially identical to those for the EPOCH model; the σ_0 values are slightly lower for the EPOCH model, but appear to converge towards the MU-SIGMA-XI model's values. Based on this analysis, we deem the MU-SIGMA-XI model to provide an accurate estimate of the GEV model parameters for all stations and all record lengths (between 40 and 2500 years).

3.3. Optimum model given 200-year records

While the MU-SIGMA-XI model may be the least biased no matter the record length, fitting to six parameters means that there is considerable uncertainty in measurement around the estimated values for each of the parameters. Can we set some of the trend terms to zero to reduce the quantified uncertainties in the model fits without introducing noticeable systematic error? In Fig. 6 we examine estimates of the 1-in-100-year rainfall amount under conditions of $T'_{SHland} = 0.0$ °C (pre-industrial), $T'_{SHland} = 1.5$ °C (slightly warmer than current), and $T'_{SHland} = 3.5$ °C (warm future) conditions for all of the possible GEV models.[1] We use 200 years of data (40 from each of the five epochs) as an approximate longest record one could hope to get from a climate model simulation running from pre-industrial conditions (about year 1900) to warm future conditions (about year 2100). For all models, quantified uncertainties are lowest at $T'_{SHland} = 1.5$ °C (green) because that is close to the average temperature across the five epochs used for the parameter estimation. We take as our nominal 'truth' the 1-in-100-year rainfall values from the MU-SIGMA-XI model using 2500 years of data (vertical lines in Fig. 6), against which to compare all the other models.

The estimated return values increase by 20% to 30% with 3.5 °C of T'_{SHland} warming according to MU-SIGMA-XI, depending upon the location. While the quantified uncertainties of the STAT model (no temperature-dependent terms) are low, it, by definition, misses the considerable dependence on temperature. Consistent with the low-bias in the σ_0 parameter, the EPOCH model matches the MU-SIGMA-XI target model with similar uncertainties but with a low bias in the return value estimates. The MU and SIGMA models also have considerable errors relative to the MU-SIGMA-XI target model.

The XI model, in which only the shape parameter depends on temperature, matches the MU-SIGMA-XI target values, but without a narrowing of quantified uncertainty despite the loss of two parameters. This model has a small number of extreme outliers (beyond the 5%–95% confidence range) and a larger spread of ξ_1 values across the bootstrap samples than for the MU-SIGMA-XI model, suggesting that the XI model is managing decent fits using the flexibility of the shape parameter but without the shape parameter being robustly representative of warming-induced changes. The MU-XI and SIGMA-XI models have small errors that might be considered tolerable, but without a reduction in quantified uncertainty (and in fact with an increase in the case of SIGMA-XI).

The MU-SIGMA model stands out for all locations in estimating the same median and mean return values as the MU-SIGMA-XI target model whilst having a substantially smaller quantified uncertainty. The reduction in uncertainty at $T'_{SHland} = 0.0$ °C and $T'_{SHland} = 3.5$ °C approaches 50%; at $T'_{SHland} = 1.5$ °C there is little difference because it is near the centre of our fitting range. We note that this aligns with the results of Brown et al. (2014), who also allowed all three parameters to vary yet concluded that it was optimal to keep the shape parameter fixed.

[1] Note that since these values are based on T'_{SHland}, strictly speaking they do not correspond exactly to global warming thresholds as stipulated in current international climate policy, but our main focus here is on understanding the sense of extreme rainfall **changes** with warming in a way that is unaffected by this difference.

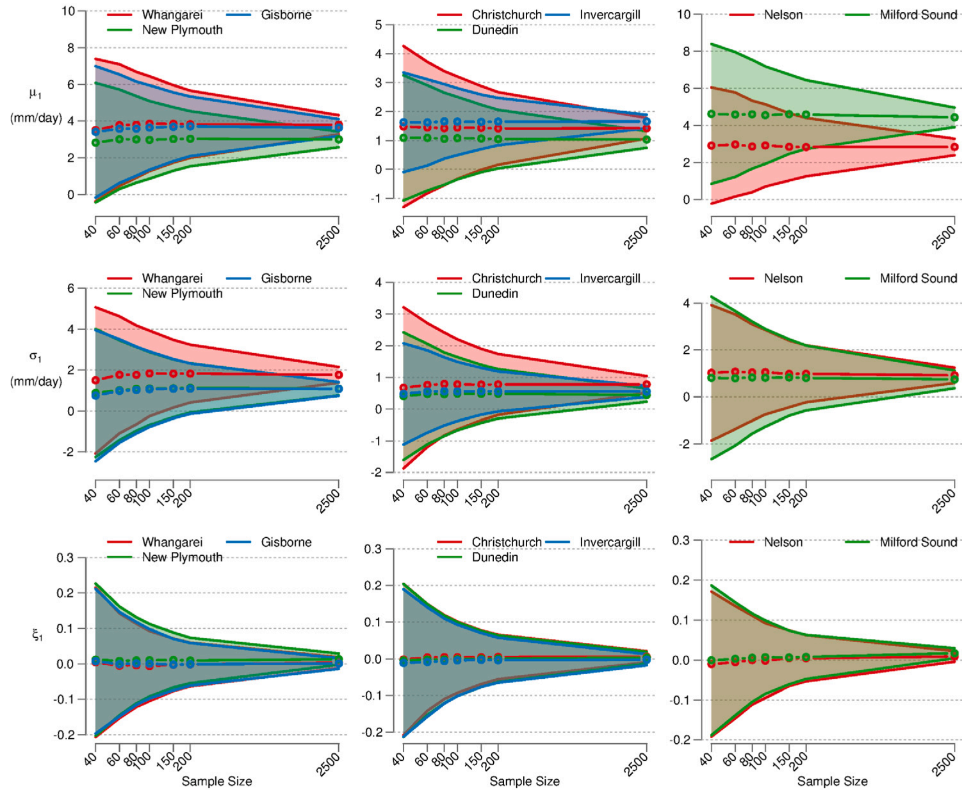

Fig. 5. The three trend GEV parameters (rows: μ_1, σ_1 and ξ_1) computed for the MU-SIGMA-XI model as a function of sample size (number of years of model data used) and location (colours, and grouped into regions in the columns). The circles indicate the mean estimate and the shading the one sigma standard error.

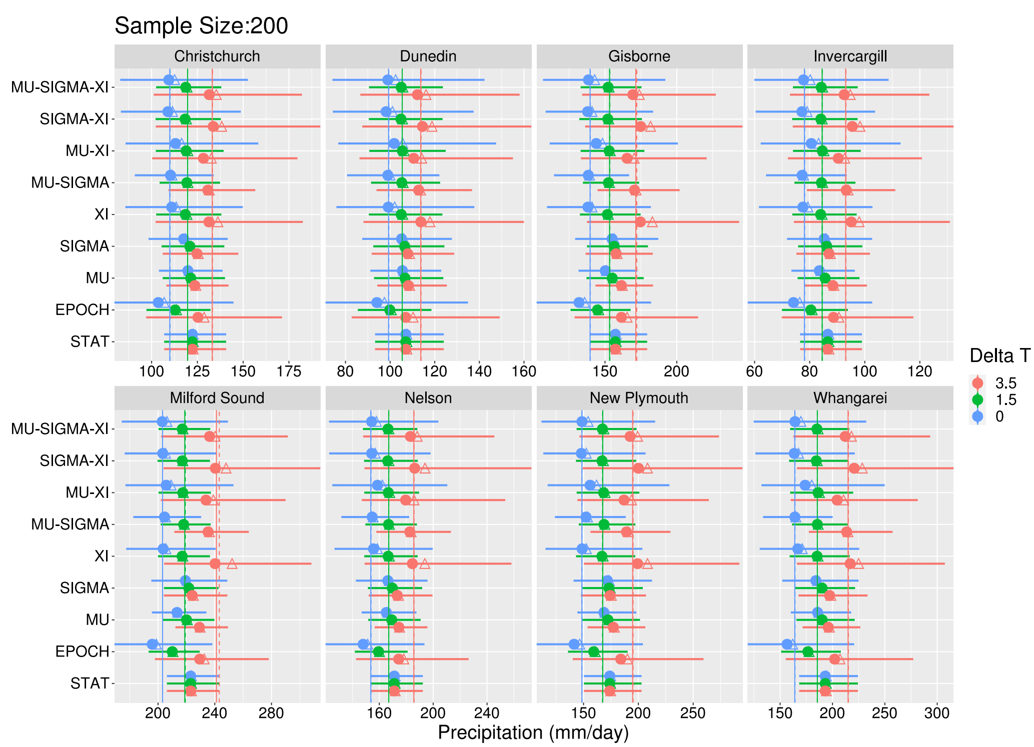

Fig. 6. The estimated 1-in-100 year daily rainfall amount (mm, across the bottom of each plot) for each of the eight locations across NZ using 200 years of climate model simulation data. Estimates are made using the nine GEV models listed in Table 1. Blue, green, and red dots show the median estimates at $T'_{SHland} = 0.0$ °C, $T'_{SHland} = 1.5$ °C, and $T'_{SHland} = 3.5$ °C of warming above pre-industrial level respectively; horizontal lines indicate the 5%–95% confidence ranges and the triangles the mean estimates. Vertical solid and dotted lines (our nominal 'truth') mark the median and mean estimates, respectively, from the fully comprehensive MU-SIGMA-XI model when using 2500 years of data. Note that the x-axis has been truncated so the full 5%–95% confidence range is not shown for all models.

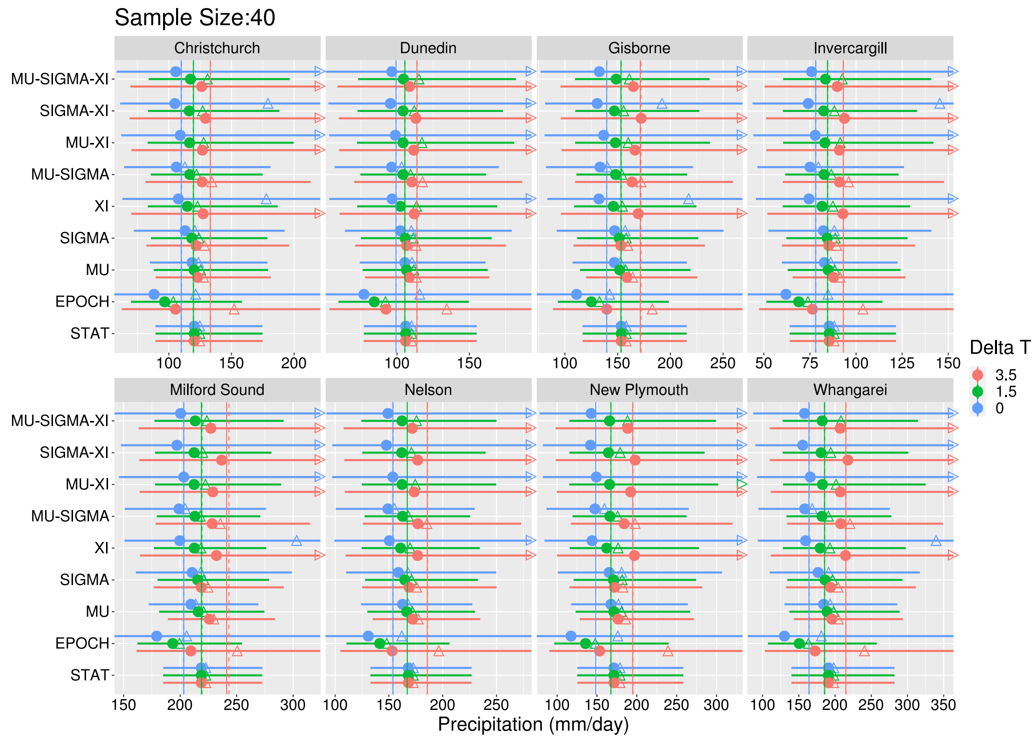

Fig. 7. The estimated 1-in-100 year daily rainfall amount (mm, across the bottom of each plot) for each of the eight locations across NZ using 40 years of climate model simulation data. Estimates are made using the nine GEV models listed in Table 1. Blue, green, and red dots show the median estimates at $T'_{SHland} = 0.0$ °C, $T'_{SHland} = 1.5$ °C, and $T'_{SHland} = 3.5$ °C of warming above pre-industrial level respectively; horizontal lines indicate the 5%–95% confidence ranges and the triangles the mean estimates. Triangles pointing left or right indicate that the means are beyond the plotting range. Vertical solid and dotted lines (our nominal 'truth') mark the median and mean estimates, respectively, from the fully comprehensive MU-SIGMA-XI model when using 2500 years of data (the same as in Fig. 6). Note that the x-axis has been truncated so the full 5%–95% confidence range is not shown for all models.

3.4. Optimum model given shorter records

Do the above conclusions hold if we are not fortunate to have 200 years of simulation or if the intended application of these GEV models is on observational records, which are typically much shorter than 200 years? Fig. 7 shows the estimated 1-in-100-year rainfall amounts when only 40 years of data are available (eight per epoch), typical of many observational records in New Zealand. We note three general differences when the record lengths are reduced from 200 years to 40 years:

- First, the quantified uncertainties for the MU-SIGMA-XI model are now huge, spanning 100% of the medians and overlapping the medians at all three T'_{SHland} values.
- Many models have a number of bootstrap cases with extremely high return values, producing a skewed tail of high values. These occur often enough to affect the 5%–95% range and are so large that the mean becomes a poor representative metric. This is particularly relevant if such a GEV model construct is intended to be applied to a 40-year observational record for which there is only a single ensemble member; the return level has a high likelihood of being significantly over-estimated.
- The low bias in the EPOCH model is exaggerated, such that the $T'_{SHland} = 3.5$ °C return value estimates actually more closely match the true (i.e. MU-SIGMA-XI) values for $T'_{SHland} = 0.0$ °C.

With this sample size, the MU-SIGMA model looks even more useful. First, the quantified uncertainties are reduced at least as much as with 200 years. Secondly, while it does produce a slight low bias in the medians, due to a slight low bias in the estimated σ_0 values (not shown), this bias is small compared to both the quantified uncertainty from limited sampling and the changes across the 3.5 °C range. Thirdly, this model does not produce outlier values that skew the distribution and affect the mean. This makes this GEV model construct particularly attractive to application to historical records where, given the absence of multiple ensemble members, it is imperative that the likelihood of over-estimating the return level is minimised.

With a 40-year sample, the MU model becomes more appealing. It has slightly smaller quantified uncertainties than the MU-SIGMA model. While it does have a systematic bias towards the value at the central temperature of the fit ($T'_{SHland} \sim 1.5$ °C), this bias might be considered an acceptable trade-off for reducing the uncertainty of measurement. This contrasts with the 200-year sample case, where the quantified uncertainties are low enough that the systematic bias is more noticeable.

3.5. Optimum model assessment using statistical metrics

Finally, our approach to selecting an optimal model for our purposes is augmented by considering more rigorous statistical metrics. The Bayesian and Akaike Information Criteria (BIC, AIC) provide metrics for assessing model fit performance, based on values of the log likelihoods associated with the fits, and including penalty terms for any extra parameters required to improve the fit. We computed BIC and AIC at all locations for all models except 'EPOCH' (see caption) and these appear in Fig. 8. First, we note that in every case almost all of the non-stationary models improve on the stationary ('STAT') approach, confirming the appropriateness of considering non-stationarity in extreme rainfall here. Generally speaking, the AIC always chooses MU-SIGMA as the best model, in alignment with our previous analysis, although the BIC usually chooses MU first and then MU-SIGMA second. This is perhaps to be expected given that the BIC penalises more heavily than AIC for the additional parameter involved in MU-SIGMA compared with MU.

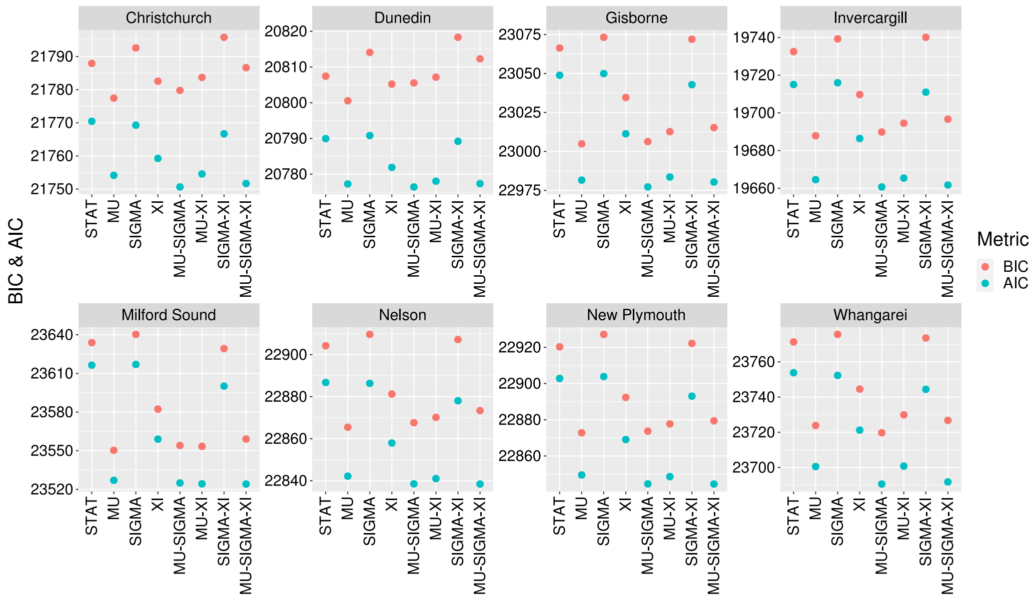

Fig. 8. Bayesian Information Criterion (BIC, red) and Akaike Information Criterion (AIC, blue) at all locations, using a 2500 sample size, for all models except 'EPOCH'. ('EPOCH' was omitted as computation of the BIC/AIC for this model would be non-trivial, given its very different treatment of non-stationarity). BIC and AIC aid in model selection by providing a measure of model performance whilst also taking account of the number of parameters used. Lower values of BIC/AIC indicate a better performing model, although BIC applies a stiffer penalty than AIC for any extra parameters used to improve performance.

We additionally assessed the quality of the non-stationary model fit by standardising the data and examining the resulting Q-Q plots, following section 6.2.3 of Coles (2001). In most cases the non-stationary fits appear good, and indicate an improvement over the 'STAT' model. There are, however, some exceptions – notably Milford Sound, New Plymouth and Nelson – where perhaps the GEV distribution is not flexible enough to describe the annual maxima series at these locations.

4. Discussion

Focusing on extreme rainfall in the New Zealand region, and basing our investigations around the GEV model, we set out to discover whether there is any particular model construct that can give acceptably accurate estimates whilst minimising the complexity of the model. Our 'MU-SIGMA' model indeed provides such an optimum. In this model the non-stationary approach allows the location and scale, but not the shape, parameter to vary linearly with a temperature covariate indicative of climate warming, in this case the Southern Hemisphere land temperature anomaly. With the location and scale parameters cast in this way, the GEV fit is performed such that both fixed and trend terms of the parameters are computed as part of the fit.

The computed parameters for the 1-in-100-year rainfall amount at the eight locations have a number of implications for understanding the shifting patterns of rainfall extremes in the NZ region (Griffiths, 2011). First and foremost, all locations have rainfall depths increasing with warming (Fig. 6), broadly in line with theoretical expectations in a warming world (Fowler et al., 2021a; Allen and Ingram, 2002; Pall et al., 2007). In this context, the inclusion of the 'STAT' model may appear at odds, since by definition it includes zero dependence on temperature; however, previous research on rainfall extremes in the region has employed the relatively short available observational record in precisely this way (Carey-Smith et al., 2018), so our inclusion of it here provides a baseline for the value of considering the anthropogenic response.

For mid-latitude regions such as New Zealand, it is generally expected that, for the rarest and shortest duration rainfall events, rainfall amounts in a warming world should increase roughly in line with increases in water vapour under invariant relative humidity, at a rate of around 7% per °C of warming (Fowler et al., 2021a,b; Lenderink and Fowler, 2017). However, this is a very rough rule of thumb that tells us only about the amount of moisture a warmer atmosphere can potentially hold, not the amount of rainfall directly (Adam, 2023). The actual changes to rainfall in a warmer atmosphere will depend on many factors, including moisture availability and the processes generating the rainfall. Systems originating over drier land areas may struggle to achieve the 7%/°C possible, in contrast to those over the ocean, with its virtually limitless moisture supply. Storm dynamics may lead to changes higher than the Clausius–Clapeyron rate by generating enough kinetic energy to entrain plentiful surrounding air, then re-releasing energy via latent heat release such as to 'super-charge' the storm (Adam, 2023; Trenberth, 1999; Liu et al., 2019; Westra et al., 2014; Pall et al., 2017).

Table 3
How precipitation extremes are changing with warming according to our most comprehensive model, MU-SIGMA-XI. The second column shows the trend component of the location parameter as a percentage of the invariant component; the third column shows the same for the scale parameter normalised by the value for the location parameter. The final column shows the resulting percentage change in precipitation amount in % per °C of warming.

Site	Location trend $100 \times \frac{\mu_1-\mu_0}{\mu_0}$	Scale trend $100 \times \frac{\sigma_1-\sigma_0}{\sigma_0} / \frac{\mu_1-\mu_0}{\mu_0}$	Percentage trend (%/°C)
Whangarei	6.5 (4.9, 8.1)	1.4 (0.9, 2.0)	8.1 (4.1, 12.1)
New Plymouth	5.7 (4.2, 7.1)	1.1 (0.5, 1.8)	8.2 (3.8, 12.9)
Gisborne	7.3 (5.5, 9.1)	0.8 (0.4, 1.2)	6.2 (2.6, 9.8)
Nelson	4.0 (2.8, 5.1)	1.3 (0.2, 2.2)	5.6 (2.4, 8.9)
Milford Sound	3.8 (1.9, 4.7)	0.5 (−0.7, 1.8)	5.2 (2.6, 7.7)
Christchurch	3.8 (2.2, 5.4)	1.5 (0.8, 2.4)	5.6 (1.9, 9.4)
Dunedin	3.1 (1.7, 4.6)	1.2 (0.2, 2.3)	4.0 (0.0, 8.1)
Invercargill	5.8 (4.4, 7.2)	1.1 (0.5, 1.6)	5.2 (1.7, 8.8)

We use the full MU-SIGMA-XI GEV model to estimate the rate at which rainfall extremes are changing with warming in the weather@home/ANZ climate model (Table 3). Estimated increases range from 4%/°C (Dunedin, East of South Island) to 8%/°C (New Plymouth, West of North Island), but the uncertainties are large enough to more than envelope this range at all locations. The higher rates in the north and northwest might be explained by extra dynamically-induced contributions from entrainment and advection of super-moist air from further north associated with atmospheric rivers (Reid et al., 2021) and also orographic enhancement from westerly winds meeting high topography. Since extreme rainfall along the eastern side of NZ requires more unusual atmospheric flow, the lower rainfall enhancement rate there suggests that there may be a climate-induced change to the likelihood of conducive flows. Milford Sound and Nelson perhaps represent slightly surprising results. Milford Sound is on the exceptionally wet West Coast, routinely exposed to very moist air that has travelled thousands of kilometres across ocean, and Nelson, at the top of the South Island, is exposed to the North, vulnerable to atmospheric rivers. As such, we might anticipate rainfall changes at both these locations to be more in line with the 7%/°C rate. We note, however, that these are two of the locations for which the non-stationary GEV fit was poorest; increasing understanding and confidence in any assessment of extreme rainfall trends in these locations in particular will no doubt require further research.

The MU-SIGMA-XI model provides some insights as to how the change in return values is materialising (Table 3). For instance, the relative climate change-induced increase in the location parameter (μ_1/μ_0) differs across the country, being substantially higher in the three northernmost locations than elsewhere. Furthermore, while uncertainties are large, the increase in the scale parameter appears to result simply from the stretching of the distribution expected with the increase in the location parameter: the relative increase in the scale parameter (σ_1/σ_0) matches the relative increase in the location parameter (μ_1/μ_0). We acknowledge, however, the substantial uncertainty around this, as is inherent in any attempt to understand the non-stationary nature of rainfall extremes. As noted by Serinaldi and Kilsby (2015), the inductive approach, making assumptions about how the fit parameters are changing with the covariate, introduces substantial uncertainty over the stationary approach and, depending on the application, the stationary approach may yet remain more useful. Here, our use of extremely large datasets and our objective of exploring rainfall changes in a general sense (as opposed to needing specific numbers for e.g. engineering applications) have permitted a useful exploratory study — but increased uncertainty remains.

5. Conclusions

Overall, our findings indicate that extreme precipitation is increasing in intensity across NZ at a rate that may matter for regulatory applications. Nevertheless, the rate of increase may differ across the country, reflecting possible differences in how anthropogenic climate change is affecting the mechanisms that produce local extreme precipitation. Determination of whether these spatial differences matter will require confirmation with further climate modelling and discussions with regulators and with parties who most comply with design-rainfall regulations. Those further climate modelling efforts can potentially benefit from another key outcome of this study, namely that the optimal approach when estimating rainfall extremes with the GEV model appears to be the non-stationary approach in which the location and scale, but not the shape, parameters are allowed to vary with climate warming. The testing of this over sample sizes much larger than available in the historical observational record is a key beneficial outcome of this study.

CRediT authorship contribution statement

Suzanne Rosier: Writing – review & editing, Investigation, Data curation, Visualization, Supervision, Project administration, Validation, Resources, Methodology, Funding acquisition, Conceptualization, Writing – original draft, Formal analysis. **Shalin Shah:** Writing – review & editing, Visualization, Software, Formal analysis. **Greg Bodeker:** Writing – review & editing, Methodology, Data curation, Software, Visualization, Conceptualization, Formal analysis. **Trevor Carey-Smith:** Formal analysis, Writing – review & editing, Software, Visualization, Investigation, Conceptualization, Methodology. **David Frame:** Funding acquisition, Conceptualization, Writing – review & editing. **Dáithí A. Stone:** Writing – original draft, Formal analysis, Methodology, Writing – review & editing, Supervision, Investigation, Conceptualization, Visualization, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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